TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 3

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

Information:

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a basic calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

Items:

- You **will need** a pen with black or blue ink, a pencil, a ruler and an eraser (for diagrams, graphs and charts only)
- You **will need** a basic calculator for Section B only
- You **will need** a protractor and a compass
- You **will not** need any other stationery or equipment.

Time allowed:

30 minutes for Section A (Non-calculator)
90 minutes for Section B (Calculator)

Do not open this examination paper until you are told to do so.

For examiner use only

	Marks available	Marks awarded	Second marks
Section A	15		
Section B	45		
Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 0.238 + 0.05

Answer	

[1]

2. Work out 0.8 × 0.222

Answer	

[1]

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3. A company has 200 employees and 2 offices.

Employees are split across both offices as shown in the table below:

	Office A	Office B
Part-time employees	64	48
Full-time employees	16	72

One person is chosen at random.

What is the probability that the person chosen works full time in office B?

Give your answer as a percentage.

Answer	**%**

[2]

4. Round 39.4528 to the nearest whole number.

Use your rounded number to **estimate** the answer to 39.4528 × 2

Show your working.

Answer	

[2]

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5. Idris sees this mobile phone in a sale:

Mobile phone
Was £95.00
Now 12% off

How much money will Idris save buying the phone in the sale?

Answer	**£**

[2]

6. Work out $10^2 – 25 \times 4$

Answer	

[2]

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7. Work out:

$$5\frac{1}{3}+4\frac{1}{5}$$

Give your answer as a mixed number.

Answer	

[2]

8. Alex has just been on a cruise:

- the cruise ship travelled 4320 miles
- it took 10 days of non-stop sailing.

Alex thinks the average speed of the cruise ship was 20 miles per hour.

Is Alex correct?

Show how you decide.

Answer	

[3]

End of Section A.

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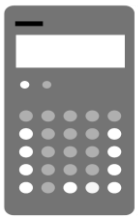
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Section B begins on Page 6.

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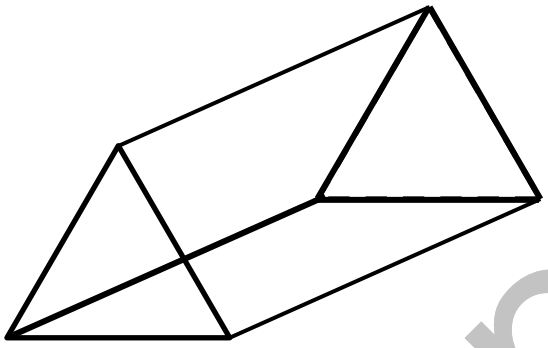
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Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. What is this shape?



Answer	

[1]

2. Put these numbers in order starting with the lowest:

1 025 600 –102 560 –85 956 1 102 560

Answer				
	Lowest	→		**Highest**

[1]

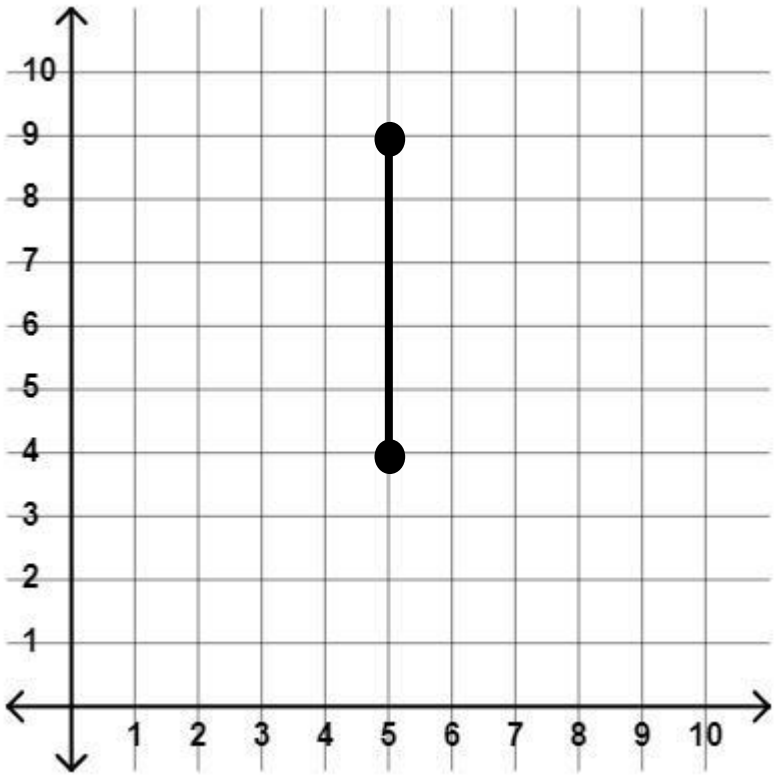
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3. Ezra is going to put a shed in a garden.

The shed is in the shape of a square.

The coordinate grid shows where two corners and one edge of the square will go.



What are the possible coordinates for the other **two** corners of the square?

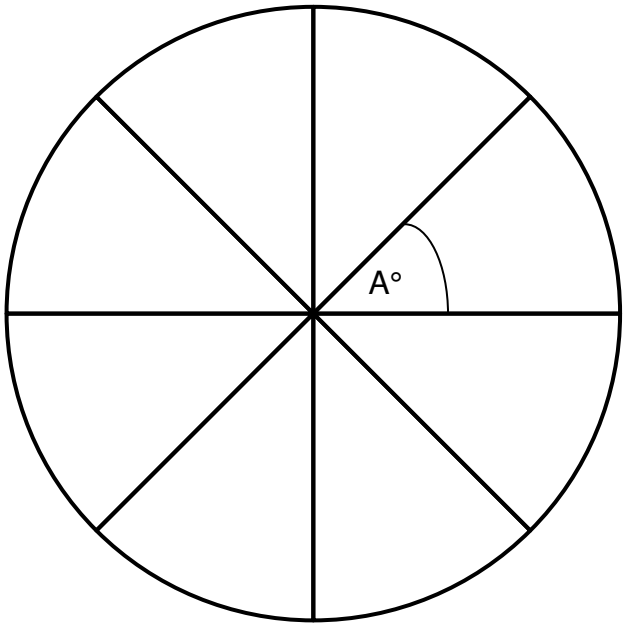
Answer	(,) and (,)

[2]

Questions continue on the following page

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4. A circle is divided into 8 equal sections as show below:



Calculate the value of A.

Answer	°

[2]

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5. Write this fraction:

$$\frac{29}{40}$$

(a) as a decimal

(b) as a percentage.

Answer	**(a) Decimal:**	**(b) Percentage:**

[2]

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Questions continue on the following page

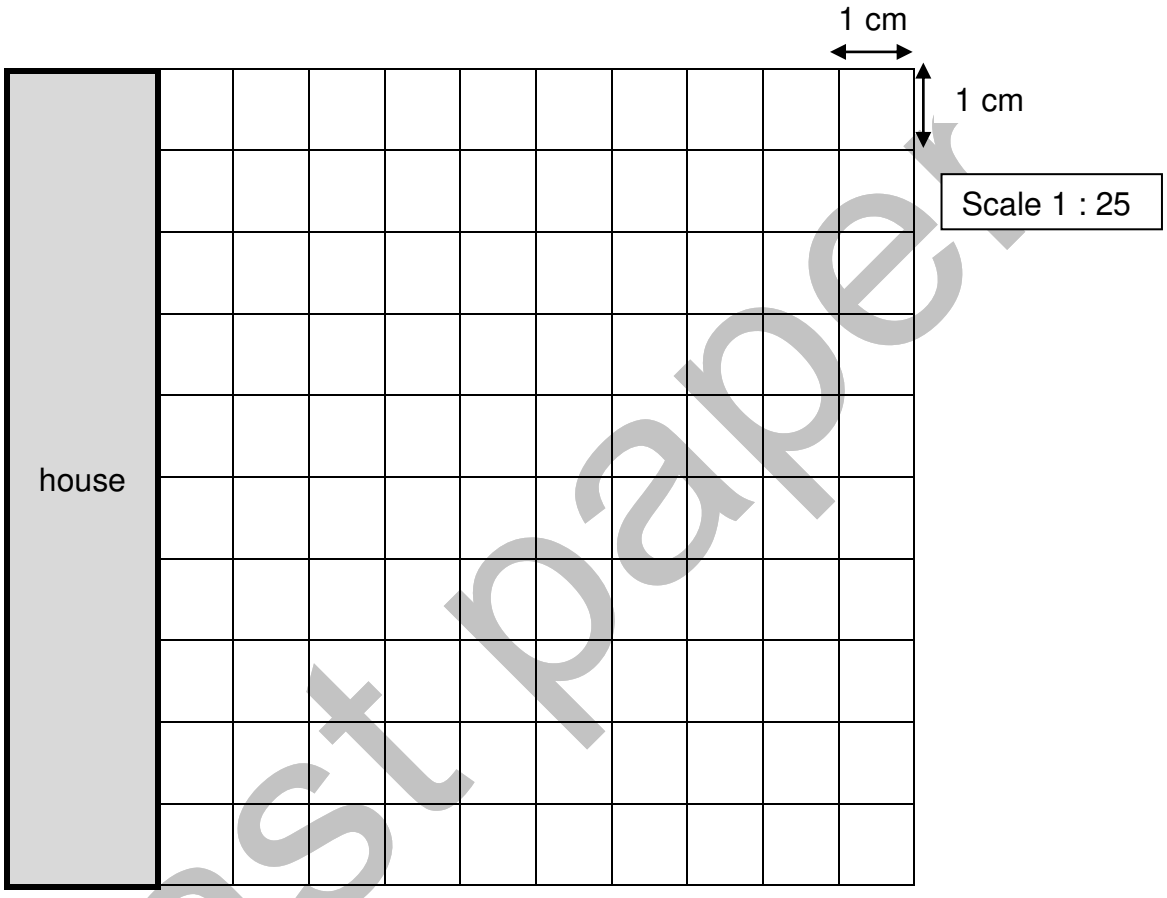
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6. Charlie wants to build a pond in a garden.

The pond needs to be a rectangle with:

- a length of 2 metres and a width of 1 metre
- **at least** 1 metre away from the house.

Draw a possible position for the pond on the scale diagram below:



[2]

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7. Last year 825 500 people visited a museum.

The manager is set a target to increase the number of visitors by 22%.

The manager thinks this will be more than 1 005 000 visitors.

Is the manager correct?

Show how you decide.

Answer	

[3]

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8. Leslie needs to put a fence around a pond.

The pond is in the shape of a circle with a diameter of 4.26 meters.

Leslie buys 13.5 metres of fencing.

Has Leslie bought enough fencing?

Show how you decide.

Use π = 3.14

Answer	

[3]

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9. Taylor books a holiday.

The next day, their friend books the same holiday for £690 **after** receiving a 4% discount.

How much more did Taylor pay for the holiday?

Answer	**£**

[3]

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10. Chen sells cakes.

The table below shows the ingredients in one cake:

Ingredient	Grams
Flour	275
Sugar	200
Baking powder	5

A customer wants to know what fraction of the cake is sugar.

Chen tells the customer $\frac{7}{12}$

Is this correct?

Show how you decide.

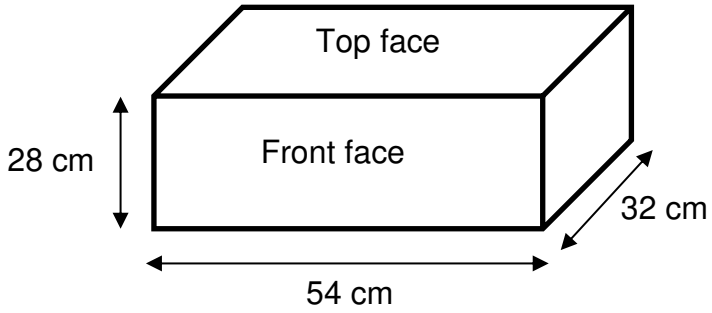
Answer	

[3]

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11. Lee wants to paint the front face and top face of this cuboid as part of an art project:



Not drawn to scale

Lee will paint yellow, green and white stripes in the ratio 5 : 3 : 2

How much of the cuboid will be painted yellow?

Give your answer in square centimetres (cm^2).

Answer	**cm^2**

[4]

Questions continue on the following page

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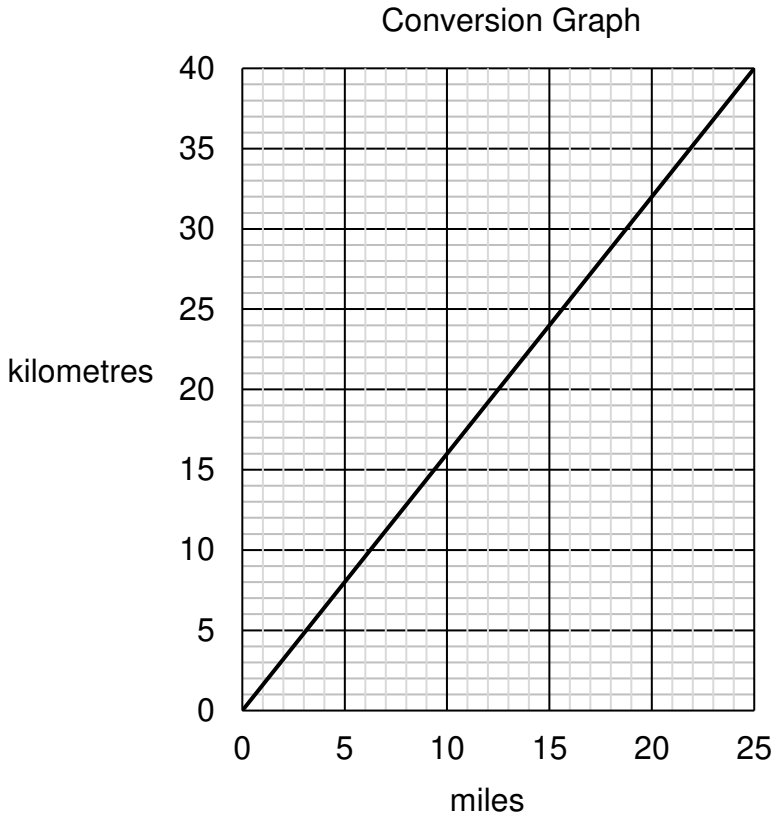
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12. Harper is planning a road trip with this information:

- the total distance is 1600 **kilometres**
- Harper only wants to drive a maximum of 225 **miles** each day.

How many whole days will Harper need to complete the 1600 km?

Use this graph to convert between kilometres and miles.



Show your working.

Answer	**days**

[4]

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13. Over a period of six weeks, Jamie sold chocolate cupcakes and lemon cupcakes.

The number of chocolate cupcakes Jamie sold each week is shown below:

Number of chocolate cupcakes sold					
Week 1	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**
91	85	64	68	100	72

The number of lemon cupcakes Jamie sold is summarised below:

Lemon cupcakes sold	
Mean number sold per week	84
Range	20

Jamie claims:

a) "On average, I have sold more chocolate cupcakes per week"

b) "The chocolate cupcake sales are more consistent".

Are both of Jamie's claims correct?

Give reasons for your answers.

Show working here:
Reason - claim a):
Reason - claim b):

[4]

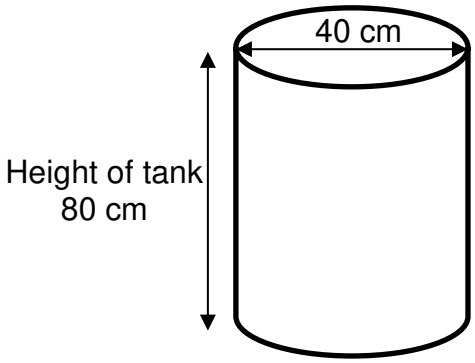
Questions continue on the following page

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14. Ali is setting up a saltwater fish tank.

The fish tank is in the shape of a cylinder.



Not drawn to scale

Ali uses this formula to work out how many grams of salt to add to the tank:

$$\text{Amount of salt in grams} = \left(\frac{\text{volume of tank}}{1000}\right) \text{multiplied by } 35$$

Ali thinks the tank will need more than 3.75 kilograms of salt.

Is Ali correct?

Show how you decide.

Use π = 3.14

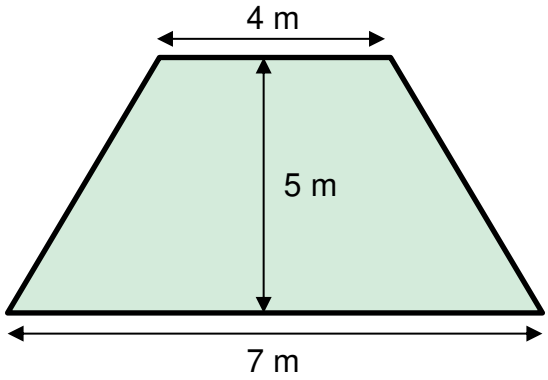
Answer	

[5]

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15. Riley wants to estimate how much it will cost to lay a patio in the garden below:



Not drawn to scale

Riley finds these prices online:

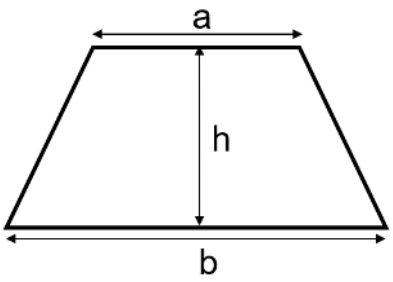
Price per square metre (m^2)					
£85	£100	£75	£130	£150	£135

Riley uses the median price to estimate how much it will cost to lay the patio.

How much does Riley estimate the patio will cost?

Use:

$$\text{Area} = \frac{(a+b)h}{2}$$



Show your working.

Answer box is on the next page

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Answer	**£**

[6]

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This is the end of the assessment.

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